

Mitigating Cumulative Privacy Risk in Continual Information Sharing: A Dynamic Stackelberg Game Approach

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Abstract

Privacy leakage on Web-based platform has become a critical challenge as users continually share personal information through online social networks, health tracking platforms, and other Web services. Information recipients and third-party can progressively aggregate shared content across the Web, enabling increasingly accurate profiling of individuals. However, existing studies typically treat each disclosure as independent, overlooking the cumulative privacy risks that arise in continual information sharing. In addition, the subjective cognition of both users and adversaries in Web environments, where users and adversaries can dynamically adapt based on observable actions, remains underexplored. To address these challenges, we propose a dynamic Stackelberg game model for continual information sharing scenarios, where user's sequential privacy decisions are optimized to balance privacy protection and data utility. The model explicitly captures the cognitive behaviors of both the user and the adversary, allowing their subjective perceptions to shape the Stackelberg equilibrium. Building on this formulation, we develop a reinforcement learning-based algorithm to derive approximately optimal strategies for mitigating privacy leakage in the context of continual information sharing. Experiments on real-world datasets demonstrate that our method significantly reduces cumulative privacy risks while preserving the utility of shared content. The proposed model further provides actionable insights for the design of privacy-enhancing technologies and web platform policies.

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CCS Concepts

• Security and privacy → Human and societal aspects of security and privacy → Privacy protections.

Keywords

Dynamic Stackelberg game, Cumulative privacy risk, Attribute inference, Continual information-sharing, Privacy protection

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1 Introduction

The privacy protection of Web services has attracted extensive attention from researchers over the past decade and continues to raise significant concerns among users. The privacy paradox [2, 7] has become nearly ubiquitous in the context of Web service usage. This phenomenon refers to the situation where users, either actively or passively, disclose personal information in order to obtain higher-quality or more personalized services at the expense of potential privacy leakage. Such cases are especially common in information-sharing scenarios, including online social networks, e-commerce platforms, and smart home systems.

As the amount of personal information shared by users increases, adversaries can exploit this data to launch privacy attacks. Among these, privacy inference attacks [4, 8, 15], which aim to infer hidden or sensitive user information, have emerged as one of the most prominent privacy threats. In particular, attribute inference attacks which designed to deduce users' concealed attributes such as location, income, sexual orientation, or behavioral habits have been widely studied in domains such as online social networks [1, 27], recommender systems[14] and query-based systems [22].

Generally, the statistical or semantic correlations between observable information and hidden attributes serve as the key basis for attribute inference. To counteract such attacks, researchers have proposed various methods for detecting and analyzing attribute inference [14], [22]. Meanwhile, several studies have explored privacy measurement under inference threats, focusing on metrics such as information gain and uncertainty [13]. On the other hand, since privacy protection typically leads to loss of data utility and may degrade service quality, the tradeoff between privacy and utility is introduced in the countermeasures against attribute inference attacks [9, 17, 27].

Most existing approaches, however, assume that each instance of information disclosure is independent, thereby overlooking the fact that users often face accumulative privacy risks arising from continual information sharing. In such settings, an adversary can enhance attribute inference performance by persistently observing or collecting shared data over time, obtaining richer correlations related to the target user's private attributes. Although several studies have explored accumulative privacy risks [17, 20, 30], they often neglect the influence of both adversarial and user cognition, as well as the underlying game-theoretic relationship between them. In practice, the adversary's cognition affects the subjective estimation of correlations between observable and private information, as well as the perceived cost and payoff of inference. Conversely, the user's cognition influences their awareness of potential privacy leakage and their personal preference toward privacy risks. The strategic game between these two parties determines how they select adversarial and defensive strategies under cognitive uncertainty.

In this paper, we focus on attribute inference in the context of continual information sharing, where adversaries and users make inference and protection decisions based on their respective cognitions within a game-theoretic framework. In such scenarios, privacy depends not only on individual disclosure events but also the accumulative effects of past disclosures, which fundamentally distinguishes continual information sharing from independent disclosure settings. Meanwhile, the adversary's cognition regarding the correlation between privacy and available information directly impacts their inference strategy. Accordingly, users must derive counter-strategies to mitigate privacy risks, where their cognition determines both their risk preferences and the strength of their protective actions.

To alleviate the impact of continual information disclosure under the cognitive interaction between adversary and user, three key challenges must be addressed:

1. In general, privacy risk progressively increases as adversaries continually observe shared information to enhance the inference performance. Therefore, effective privacy protection requires imposing constraints on each disclosure action to limit accumulative risk.
2. The dynamic interaction between adversaries and users shapes the overall effectiveness of privacy protection. Since inference threats arise after information disclosure, accurately characterizing the risks associated with potential adversarial inference strategies is crucial for deriving optimal privacy policies.
3. The cognitions of both adversaries and users introduce uncertainty into their decision-making processes. Enhancing the adaptability of privacy protection thus depends on quantifying how

cognition influences strategic choices and on developing optimal responses based on inferred opponent cognition.

To address above challenges, we propose a Stackelberg game-based model to characterize the sequential interaction between users and adversaries in continual information sharing. By modeling the effects of previous disclosures and incorporating cognitive factors into the game, we derive the Stackelberg equilibrium that captures the optimal strategies for both players. Moreover, we design a reinforcement learning-based algorithm to approximate the user's optimal strategy efficiently. The main contributions of this study are as follows:

1. We resist the attribute inference attack in a practical continual information sharing scenario, overcoming the common assumption of independent disclosures. This enables stronger privacy protection for cases involving highly sensitive information sharing.
2. We analyze the cognitive impact of both adversaries and users on decision-making. By integrating these effects into a Stackelberg game, our model enhances adaptability to various adversarial capabilities and user privacy preferences.
3. We prove the existence of a Stackelberg equilibrium and derive the optimal strategies for both adversaries and users. Furthermore, we propose an efficient reinforcement learning-based algorithm for users to approximate the optimal strategy, achieving a practical tradeoff between privacy and utility.

The rest of the paper is organized as follows. We review some related work in Section 2 and formulate the problem in Section 3. We then present the solution in Section 4 and report the experimental results in Section 5. Finally, we conclude the paper in Section 6 in which we also discuss the direction of our future work.

2 Related Work

Privacy inference. Privacy inference refers to the process in which an adversary infers hidden information about a target user from related observable data. This topic has been widely studied in areas of machine learning [10] and information sharing [23]. Wang et al. [25] proposed a visual model to illustrate and explain the underlying inference behaviors via a Bayesian Network, which could visually explain the process details of the underlying privacy preserving models. Yi et al. [28] investigated privacy inference in online social networks and proposed a privacy-sensitive data identification model to recognize the data that are critical to users' sensitive data. To analyze the privacy inference threats posed by large language models, Yukhymenko et al. [29] constructed a diverse synthetic dataset of over 7800 comments manually labeled for personal attribute, thereby partially addressing the lack of public datasets for this task. Stevanoski et al. [22] introduced QueryCheetah, a model designed to detect attribute inference attacks against Query-based system, where the attribute inference process is formulated as a privacy game. Yi et al. [27] developed an evolutionary game-based framework to defend against attribute inference attacks in online social networks arising from privacy decision conflicts among multi-individuals. Based on this framework, users could cooperatively make privacy decisions to mitigate interdependent privacy risks.

Compressive privacy. Compressive privacy [9] aims to preserve data utility while protecting sensitive information, by transforming or compressing data according to known utility and privacy models. To protect the private states while maintaining public states in the linear dynamical system, Song et al. [21] proposed an optimization algorithm that identifies an optimal compression matrix and decomposes the global optimization into several local subproblems solvable at each sensor. Wang et al. [24] proposed a compressive privacy federated learning scheme to protect the task-irrelevant information from gradient leakage attacks, in where clients train a local compressive model to extracts task-relevant and removes task-irrelevant information from clients' data. Gao et al. [6] investigated privacy leakage in vehicle intelligence systems and proposed a compression matrix-based method to ensure privacy protection while preserving system functionality. Their algorithm linearly transforms high-dimensional sensor data into a lower-dimensional space at each time step before transmission to a fusion center, achieving both privacy and efficiency.

Privacy for continual information sharing. Time series data that ordered sequences of finite-length observations have attracted significant research interest in the last decade [12]. Shao et al. [16] proposed a location inference attack framework that simultaneously reconstructs multiple trajectories from differentially private trajectory data using the structured sparsity model. Dong et al. [5] designed continual observation mechanisms for fundamental functions under user-level differential privacy, without requiring prior restrictions on the data distribution. In the context of information sharing, Sharma et al. [17] proposed a privacy model that enables users to optimally allocate their data-sharing behaviors to manage privacy leakage while maximizing utility. This line of research highlights the necessity of balancing continual information disclosure with dynamic privacy protection mechanisms.

3 Problem Formulation

3.1 System Model

In this study, we consider a scenario where the privacy-aware user continually discloses personal information in exchange for certain benefits. Typical examples include users posting content on online social networks to gain social engagement, or patients sharing physiological data collected by smart medical devices with doctors for medical treatment.

Let the user's shared information at time step t be denoted as x_t . The sequence of information disclosures over time forms a continual information sharing process, $X = \{x_1, x_2, \dots, x_T\}$, where T denotes the number of sharing rounds. In each round, information recipients can observe x_t , while the user enjoys the associated utility at the cost of potential privacy leakage that may accumulate over time.

Without loss of generality, each piece of information x can be decomposed into private and public components, i.e., $x = (x^s, x^p)$, where x^s denotes the user's private information and x^p denotes the public information. These two components are often correlated, enabling adversaries to infer private information from the observable data. In practice, users may manipulate or conceal parts of their original information to mitigate privacy disclosure while maintaining acceptable utility based on their subjective perception of privacy risk. This idea can generally be realized using compressive

privacy mechanisms [9], such as query limitation, subspace noise suppression, or feature selection. The definition of compressive privacy formalized in the Definition 1.

Definition 1 (Compressive Privacy): A randomized compression mechanism $\mathcal{M} : x \rightarrow y$ is said to satisfy (ϵ, δ) -compressive privacy if, for any adversary $\mathcal{A}$, the optimal inference accuracy of a sensitive attribute $S = \text{Infer}(X)$ given the compressed observation $Y = \mathcal{M}(X)$ satisfies $\Pr[\mathcal{A}(Y) = S] \leq \Pr[\mathcal{A}_0 = S] + \epsilon$ with probability at least $1 - \delta$, where $\mathcal{A}_0$ denotes a random guess based on prior knowledge.

This definition ensures that no adversary can exceed the baseline accuracy by more than ϵ , under the success probability constraint $1 - \delta$. Following this definition, we implement the compression mechanism $\mathcal{M}$ using a compression matrix C_a parameterized by the compression strength a . The resulting compressed information can be expressed as $\hat{x} = C_a(x)$.

In continual information sharing, the data disclosed at current time step t after applying the compressive privacy mechanism can be denoted as $\hat{x}_t$. Accordingly, all the information shared by the user up to time t is represented as $\hat{X}_t = \{\hat{x}_1, \hat{x}_2, \dots, \hat{x}_{t-1}, \hat{x}_t\}$. Meanwhile, service providers or information recipients can improve the quality of service or accomplish other purpose based on the accumulative shared information $\hat{X}_t$.

3.2 Threat Model

In the continual information-sharing scenario discussed above, we assume that the privacy threats arise from semi-trust service providers and malicious information recipients. In general, users are required to exchange personal information with service providers to obtain better service quality. For example, sharing more profile details and browsing histories often enhances the performance of recommendation system. However, semi-trust service providers, although honest in service delivery, are often curious about users' private information, exposing users to accumulated privacy risks as information is continually shared over time. Similarly, sharing information with malicious recipients can also compromise users' privacy, e.g., by enabling them to infer private attributes, sell sensitive data, or further disclose information to third parties. Overall, whether it is a service provider or a malicious recipient, adversaries are not only interested in the explicit content of shared information but also tend to infer hidden user attributes from it. Therefore, the adversarial entities considered in this work include both semi-trusted service providers and malicious recipients, each capable of performing attribute inference attacks against users.

Under the above privacy threat, users typically disclose the privacy-processed data $\hat{x}_t$ at time t , instead of the original data x_t , to protect privacy through compressive privacy mechanism. Although adversaries cannot directly access the private component $x_t^s \in x_t$, they can still infer x_t^s by exploiting correlation within the accumulative information $\hat{X}_t = \{\hat{x}_1, \hat{x}_2, \dots, \hat{x}_{t-1}, \hat{x}_t\}$ along with their background knowledge $\mathcal{B}$. Without loss of generality, we formally define private attribute inference and privacy inference risk under continual information sharing in Definition 2 and 3, respectively. The diagram illustrating this privacy threat model is shown in Figure 1.

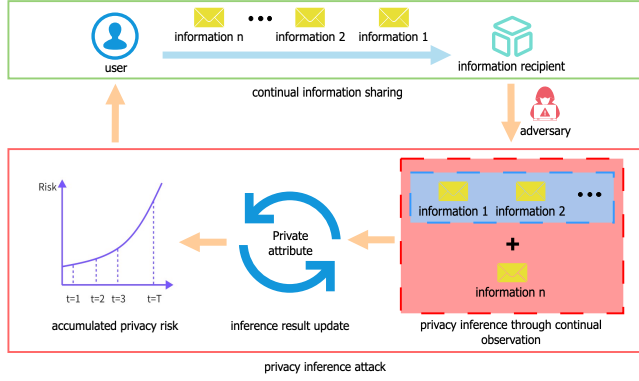


Figure 1: Privacy Inference in Continual Information Sharing

Definition 2 (Private Attribute Inference under Continual Information Sharing). Let the sequence of shared information up to time t be $\hat{X}_t = \{\hat{x}_1, \hat{x}_2, \dots, \hat{x}_{t-1}, \hat{x}_t\}$, which is observable to an adversary together with its background knowledge $\mathcal{B}$, the private attribute inference under continual information sharing indicates that the adversary could update the inference result based on the newly disclosed information, formulated as $s_t \leftarrow \text{infer}(\hat{X}_t, \mathcal{B})$.

Definition 3 (Privacy Inference Risk under Continual Information Sharing). The privacy inference risk $\mathcal{R}(s_t)$ with respect to time t is defined as a non-decreasing function of the statistical dependency between the inferred attribute s_t and the accumulative disclosure $\hat{X}_t$:

$$\mathcal{R}(s_t) \triangleq f\left(D\left(P_{s_t, \hat{X}_t} \| P_{s_t} P_{\hat{X}_t}\right)\right) \quad (1)$$

where $D(\cdot \| \cdot)$ denotes a dependence measure such as mutual information, correlation coefficient, or f-divergence, and $f(\cdot)$ is a monotone increasing mapping.

The equation (1) captures the accumulative nature of continual information sharing by ensuring that $\mathcal{R}(s_t)$ is non-decreasing whenever the added disclosures increase the dependency between s_t and $\hat{X}_t$.

3.3 Dynamic Stackelberg Game Formulation

In continual information sharing, data are repeatedly released over time, and both the user and the adversary update their strategies dynamically. The user's objective is to maximize the accumulative utility of shared data while constraining the growth of privacy leakage across time steps $t = 1, 2, \dots, T$. We model the user's decision as selecting a compression matrix $C_a \in C$ with strength a , representing a specific privacy level in compressive privacy.

Assumption 1: At each time t , the user selects compression matrix $C_a \in C = \{C_1, C_2, \dots, C_n\}$, corresponding to a privacy protection strength a .

Assumption 2: The privacy strategy C_a affects both the public and private components of information $x_t = (x_t^p, x_t^s)$. A stronger strategy a causes greater loss in data utility: $a \propto \Delta U_x$.

Given these assumptions, the user selects the privacy strategy according to her subjective privacy preference and the anticipated privacy threat. Because privacy protection and attribute inference are inherently conflicting, and because the adversary launches the

inference attack only after receiving the shared data, this interaction can naturally be modeled as a Stackelberg game. The user acts as the leader who takes an action first, and the adversary acts as the follower who responds based on the user's chosen strategy. This Stackelberg game repeats whenever the user shares new information, thus forming a multi-stage Stackelberg decision process.

Stage I: User's Strategy. Given the information x_t , the user selects C_a with protection strength a to obtain $\hat{x}_t = C_a(x_t)$, then shares it to achieve corresponding utility U_u . For a predicted strategy $b \in B$ chosen by the adversary adv , the user will select an optimal response $C_a^* \in C$ that maximizes her data utility as well as satisfies her privacy requirements:

$$C_a^* = \operatorname{argmax}_{C_a \in C} U_u(\hat{x}_t = C_a(x_t), b) \quad (2)$$

For convenience, a is used to denote C_a , since the compression strength characterizes the strategy.

Stage II: Adversary's Strategy. After receiving $\hat{x}_t = C_a(x_t)$, the adversary adv chooses the optimal strategy $b^* \in B$ that maximizes its utility U_{adv} :

$$b^* = \operatorname{argmax}_{b \in B} U_{adv}(b, \hat{x}_t = C_a(x_t)) \quad (3)$$

The Stackelberg game is solved via backward induction [19]. First, the best response of the adversary is determined for any user strategy, then the user selects a^* to maximize her own utility U_u . Therefore, the equilibrium of the Stackelberg game can be denoted by the pair (a^*, b^*) such that:

$$\begin{cases} U_u(a^*, b^*) \geq U_u(a, b) \\ U_{adv}(a^*, b^*) \geq U_{adv}(a, b) \end{cases} \quad (4)$$

In continual information sharing, the privacy risk of user accumulates as more information are shared, according to Definition 3. Therefore, the overall objective of user can be formulated as:

$$\max_{\pi_u} \mathbb{E} \left[\sum_{t=0}^T U_u(s_t, a_t, b_t) \right] \quad (5)$$

where s_t indicates the state of privacy risk at time t .

Conversely, we assume a greedy adversary who maximizes the immediate utility at each step. That is, adversary will use all previously disclosed information as available information for attribute inference. This assumption represents a worst-case adversarial scenario, ensuring robustness of the user's privacy strategy. Under this assumption, the objective of adversary can be formulated as:

$$\max U_{adv}(s_t, a_t, b_t) \quad (6)$$

4 Equilibrium Analysis and Strategy Derivation

4.1 Utility of Adversary and User

4.1.1 Utility of Adversary. Since the content of previously shared information $\hat{X}_{t-1}$ before current time t has already become a fait accompli, the user can only adjust the current disclosure x_t through choosing an appropriate privacy strategy to mitigate the accumulative privacy risk. We denote by $H_\infty(x^s)$ the prior min-entropy of the sensitive attribute x^s , representing the adversary's minimum uncertainty before observing $\hat{x}_t$, and by $H_\infty(x^s | \hat{x}_t)$ the posterior min-entropy after observing $\hat{x}_t$. Hence, the adversary's attribute

inference gain obtained from the newly observed disclosure can be expressed as the difference between these two quantities:

$$L(\hat{x}_t) = H_{t-1}^\infty(x^s) - H_t^\infty(x^s | \hat{x}_t) \quad (7)$$

Note that the adversary can only observe $\hat{x}_t = (\hat{x}^p, \hat{x}^s)$ rather than original information $x_t = (x^p, x^s)$, due to the privacy strategy a taken by user. Therefore, adversary may utilize the correlation between x^s and $\hat{x}_t$ based on its cognition instead of the true correlation $P(x^s, x_t)$ in the attribute inference. Meanwhile, the adversary's subjective cognition may impact the estimation of the correlation between x^s and $\hat{x}_t$. To account for this impact, we incorporate the estimated correlation into the formulation of the adversary's utility. This enables us to obtain a more comprehensive utility from the adversary's perspective. Let $Q(x^s | \hat{x}_t)$ denotes the adversary's estimated correlation between x^s and $\hat{x}_t$, the attribute inference gain after observing $\hat{x}_t$ can then be formulated as follows:

$$L(\hat{x}_t) = -\log \max_{x \in X} P(x^s) + \log \max_{x \in X} Q(x^s | \hat{x}_t) \quad (8)$$

Based on the equation (8), we can find that adversary aims to minimize the uncertainty of $Q(x^s | \hat{x}_t)$, i.e., to reduce $H_t^\infty(x^s | \hat{x}_t)$ for better performance of attribute inference attack. However, decreasing the uncertainty of $Q(x^s | \hat{x}_t)$ requires additional cognitive effort. Following the principle of rational inattention [13], this effort is modeled as a mental cost M_{adv} , represented by the Kullback-Leibler (KL) divergence between the true prior $P(x^s)$ and the estimated posterior $Q(x^s | \hat{x}_t)$:

$$M_{adv} = D_{KL}(P||Q) = \sum_{x^s} P(x^s) \cdot \log \frac{P(x^s)}{Q(x^s | \hat{x}_t)} \quad (9)$$

In addition to inference gain and mental cost, the punishment of attribute inference also needs to be considered in the utility of adversary. Because privacy breach is illegal and once exposed will be punished in practice. Generally, the punishment severity grows with the degree of privacy violation. Moreover, users with stronger privacy awareness are more likely to detect and report such violations. In this paper, we use the protection strength a to indicate the privacy awareness of user. A higher privacy strategy strength indicates that the user has a higher level of privacy awareness and expects better performance in terms of privacy protection. Thus, given the maximum values a_{max} of a and b_{max} of b , the punishment for privacy inference can be expressed as follows:

$$P = \frac{a \cdot b}{a_{max} \cdot b_{max}} F \quad (10)$$

where b represents the strength of attribute inference taken by adversary and F denotes the maximum punishment severity defined by regulations.

Accordingly, the overall utility U_{adv} of adversary launching attribute inference after observing $\hat{x}_t$ consists of the benefit of privacy inference $L(\hat{x}_t)$, mental cost M_{adv} and potential punishment P . Consequently, U_{adv} can be formally described as follows according to the relationships between the above elements.

$$U_{adv} = -\log \max_{x \in X} P(x^s) + \log \max_{x \in X} Q(x^s | \hat{x}_t) - \sum_{x^s} P(x^s) \cdot \log \frac{P(x^s)}{Q(x^s | \hat{x}_t)} - \frac{a \cdot b}{a_{max} \cdot b_{max}} F \quad (11)$$

Generally, the adversary seeks to find the optimal inference strength b that maximizes U_{adv} . Since the estimated posterior distribution $Q(x^s | \hat{x}_t)$ depends jointly on the user's privacy strength a and the adversary's inference strength b , we instantiate it as:

$$Q(x^s | a, b) = Q(x^s | \hat{x}_t) = 1 - e^{-\frac{\lambda b}{a + \epsilon}} \quad (12)$$

where λ is a sensitivity parameter and ϵ is a small constant to ensure numerical stability.

4.1.2 Utility of User. User aims to maximize the utility of information sharing while protecting privacy. Since the types of information-sharing utility vary and are partly impacted by subjective perceptions, this study focuses only on the effect of privacy strategies on the utility of data sharing. Meanwhile, based on past experience, the user may form expectations about the adversary's behavior in advance. Therefore, given the expected adversarial behavior, the user's utility consists of the cost of privacy leakage resulting from privacy inference as well as the reduction in data utility due to the privacy strategy. Specifically, the cost of privacy leakage depends not only on the adversary's inference strategy b but also on the user's privacy strategy a . It is therefore formulated as:

$$L_p = e^{-\frac{a}{b+1}} \quad (13)$$

On the other hand, adopting a stronger privacy strategy generally reduces data utility due to the inherent trade-off between privacy protection and data utility. Therefore, we use $I_u = -\alpha a$ to represent the impact of privacy strategy on information utility gain, where a larger a indicates stronger privacy protection and lower information utility gain, and α denotes the sensitivity coefficient of utility loss. After applying a smoothing transformation to L_p to ensure differentiability and numerical stability, the overall utility of user when sharing information with the privacy strategy a can be expressed as:

$$U_u(a) = I_u - L_p = -\alpha a - \log \left(1 + e^{-\frac{a}{b+1}} \right) \quad (14)$$

4.2 Existence of Stackelberg Equilibrium

According to the sequential interaction between the adversary and the user, as well as their respective utility, we next analyze the existence of Stackelberg equilibrium. Since the adversary's inference strategy is characterized by the estimation of the distribution of private attribute $Q(x^s | \hat{x}_t)$, the action space of the adversary can be regarded as a compact and convex set.

To extend the single-stage model introduced in the previous section, we now consider a dynamic multi-stage Stackelberg game where the state evolves as $s_{t+1} \sim P(\cdot | s_t, a_t, b_t)$. This formulation captures the long-term interactions between the user and the adversary under evolving privacy and inference contexts. Therefore, we have

Lemma 1 (Existence of Adversary's Optimal Response). Given the adversary's action space $\mathbb{B}$ and utility function U_{adv} , the adversary's optimal response to each privacy strategy a chosen by the user is defined as $b^*(a) = \arg\max_{b \in \mathbb{B}} U_{adv}(b; a)$. Under the assumptions that $\mathbb{B}$ is non-empty, compact, and convex, and that $U_{adv}(b; a)$ is continuous in both arguments and quasi-concave in b , the optimal

response correspondence $b^*(a)$ is non-empty, convex-valued, and upper semi-continuous. $\mathbb{B}^*(a)$ is non-empty and upper semi-continuous.

Proof. See the proof in Appendix A.

Theorem 1 (Existence of Dynamic Stackelberg Equilibrium). Consider a multi-stage Stackelberg game with a compact state space $s \in \mathbb{S}$, user's action space $a \in \mathbb{A}$, and adversary's action space $b \in \mathbb{B}$, all of which are non-empty, compact, and convex. Suppose the user's utility function $U_u(s, a, b)$ is continuous in (s, a, b) and concave in a . Under these assumptions, there exists a value function $V_u^*(s)$ satisfying the Bellman-type fixed-point equation:

$$V_u^*(s) = \max \left\{ U_u(s, a, b(s, a)) + \gamma \int V_u(s') P(ds' | s, a, b(s, a)) \right\} \forall s \in \mathbb{S} \quad (15)$$

where $b(s, a) \in \mathbb{B}(s, a)$ denotes a measurable selection from the adversary's best responses correspondence $\mathbb{B}^*(a) = \operatorname{argmax}_{b \in \mathbb{B}} U_{adv}(b; a)$.

Furthermore, the user's optimal policy $\pi_u : \mathbb{S} \rightarrow \mathbb{A}$ satisfies:

$$\pi_u(s) \in \operatorname{argmax}_{a \in \mathbb{A}} \left\{ U_u(s, a, b(s, a)) + \gamma \int V_u(s') P(ds' | s, a, b(s, a)) \right\} \quad (16)$$

and together with the adversary's policy $\pi_{adv}(s, a) = b^*(s, a)$, they form a dynamic Stackelberg equilibrium.

Proof. See the proof in Appendix B.

4.3 Optimal Strategy of Adversary and User

4.3.1 Optimal Strategy. Based on the utilities of both the adversary and the user, we derive the optimal strategies for each player in the dynamic Stackelberg game.

Theorem 2 (Optimal Strategy of Adversary). In the second stage, for any given strength of strategy strength a adopted by the user, the adversary's optimal privacy inference strength is:

$$b^* = \frac{a + \epsilon}{\lambda} \ln \frac{2\lambda a + F(a + \lambda)}{F(a + \lambda)} \quad (17)$$

Proof. See the proof in Appendix C.

From Theorem 2, it is evident that the optimal privacy inference strategy b^* is jointly determined by the user's privacy strength a and the severity of punishment F . Since F is typically predefined by regulatory policies, it can be regarded as an exogenous constant in this model. Therefore, once the user's optimal privacy strategy is established, the adversary can directly compute the corresponding best response b^* .

This property implies a monotonic dependency between the defensive intensity and the inference effort: a stronger privacy strategy discourages aggressive inference behaviors by increasing the marginal cost of privacy violation.

Theorem 3 (Optimal Strategy of User). Given the optimal inference strength b^* adopted by the adversary, the two-stage Stackelberg game admits a Stackelberg equilibrium. The optimal privacy strategy of user is:

$$a^* = (b + 1) \left(\ln \left(\frac{1}{b + 1} - \alpha \right) - \ln \alpha \right) \quad (18)$$

Proof. See the proof in Appendix D.

From Theorem 3, the user's optimal strength of a^* can be directly determined. The adversary can then compute her optimal inference strength b^* based on a^* . Therefore, the strategy pair $\langle a^*, b^* \rangle$ forms a Stackelberg equilibrium, ensuring that the utilities of both

the user and the adversary are maximized simultaneously. Moreover, the equilibrium pair $\langle a^*, b^* \rangle$ represents the most probable strategies adopted by rational agents under bounded rationality and regulatory constraints, providing a theoretical foundation for the predictability and stability of privacy-inference interactions in continual information-sharing environments.

4.3.2 Approximately Optimal Strategy of User. Heretofore, the utilities of the adversary and the user have been formulated, and the existence of the Stackelberg equilibrium has been further analyzed based on these utilities. However, deriving the exact optimal strategy for the user in continual information sharing remains nontrivial. First, the state space s_t and action space a_t are typically high dimensional and continuous, making closed-form analytical solutions extremely difficult or even infeasible. Second, the state transitions are affected by stochastic perturbations and cognitive biases of both the user and the adversary, which require modeling under uncertainty to ensure the robustness and adaptability of strategies in dynamic environments.

Algorithm 1: Approximately Optimal Strategy of User

Input: state space S , action space A , discount factor $\gamma \in (0, 1)$, clip parameter ϵ , learning rates α_u, α_{adv}

Output: trained user policy $\pi_u^{\theta^*}(a|s)$

1. Initialize Actor network $\pi_u^{\theta}(a|s)$ and Critic network $V_{\phi}(s)$ with parameters θ and ϕ , respectively.
 2. **for** each iteration $k = 1, 2, \dots, K$ **do**
 3. **for** each episode **do**
 4. Reset environment to initial state s_0
 5. **for** $t = 0, \dots, T$ **do**
 6. User samples action $a_t \sim \pi_u^{\theta}(a|s_t)$
 7. Adversary computes best response $b = b^*(s_t, a_t)$
 8. Environment returns next state s_{t+1} and user reward U_u^t
 9. Store $(s_t, a_t, U_u^t, s_{t+1}, \pi_u^{\theta}(a|s_t))$
 10. Compute $\delta_t = U_u^t + \gamma V_{\phi}(s_{t+1}) - V_{\phi}(s_t)$
 11. Compute $A_t = \text{GAE}(\delta_t) = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l}$
 12. Maximize the clipped surrogate objective:
 $\mathcal{L}_{CLIP}(\theta) = \mathbb{E}_t [\min(U_u^t(\theta)A_t, \text{clip}(U_u^t(\theta), 1 - \epsilon, 1 + \epsilon)A_t)]$
 where $U_u^t(\theta) = \frac{\pi_{\theta}(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$
 13. Update $\theta \leftarrow \theta + \alpha_{\theta} \nabla_{\theta} \mathcal{L}_{CLIP}(\theta)$
 14. Minimize the squared error $\mathcal{L}^V(\phi) = \mathbb{E}_t [V_{\phi}(s_t) - (A_t + V_{\phi}(s_t))]^2$
 15. Update $\phi \leftarrow \phi - \alpha_{adv} \nabla_{\phi} \mathcal{L}^V(\phi)$
 16. Set $\theta_{old} \leftarrow \theta$ and repeat until convergence or user's average utility no longer improves
 17. **end for**
 18. **end for**
 19. **end for**
-

To address these issues, we adopt a reinforcement learning-based approximation framework that allows the user to learn a near-optimal policy π_u^{θ} parameterized by θ , which maximizes the expected discounted accumulative utility:

$$\max_{\pi_u^\theta} \mathbb{E}_{s \sim \pi_u^\theta, b(\cdot)} \left[\sum_{t=0}^T \gamma^t U_u(s_t, a_t, b_t^*(s_t, a_t)) \right] \quad (19)$$

where γ is the discount factor controlling the balance between immediate and long-term rewards.

Specifically, the user's sequential decision process is modeled as a dynamic Stackelberg game, where the adversary acts as a stage-wise follower providing the best response at each step. The environment embeds this mapping into the transition dynamics, allowing end-to-end training. A Proximal Policy Optimization (PPO)-based actor-critic framework is used to train the user's policy network, iteratively updating parameters θ to maximize the objective while limiting policy divergence for stability and convergence. This approach enables the user to approximate the dynamic Stackelberg equilibrium through interaction with the environment without solving the recursive Bellman equation. The procedure is summarized in Algorithm 1, where the user only trains the PPO policy network, avoiding analytical equilibrium computation.

5 Experiments

5.1 Experiment Setting

In experiments, the privacy inference models based on Bayesian and Long Short-Term Memory (LSTM) are employed as the reference models to estimate privacy risks during continual information sharing. The Bayesian algorithm naturally fits this context, as its likelihood updates with the accumulation of observable information, making the posterior distribution increasingly concentrated. The LSTM captures correlations among features, thereby representing a more powerful adversary. The details of experimental setup, including datasets, baselines, and evaluation metrics, are described as follows.

Datasets. The Adult¹ and California Housing² datasets are used to analyze the performance the proposed model. Specifically, the feature sequences of entities are progressively revealed to the inference models, reflecting how a user's privacy risk evolves as the adversary accumulates observable information over time in continual sharing settings.

Baselines. Two state-of-the-art continual information sharing methods proposed in [17] are used as baselines for comparison:

Optimistic algorithm. This approach assumes that users expect to transition into the most favorable state at every time step, consistent with the behavior of privacy-unaware users. It optimizes both the information-sharing utility and the corresponding privacy strategy under the assumption that the next transition leads to the state with the highest expected utility.

Maximum disclosure algorithm. In contrast, this method models privacy-aware users who rationally choose their privacy strategy for each sharing event. Consequently, it adopts more conservative disclosure behaviors than the optimistic algorithm and seeks to maximize instantaneous utility while maintaining acceptable privacy levels.

Measurements. Privacy risk is quantified by the inference accuracy of the adversary, which indicates the probability of correctly inferring the user's private attributes:

$$A_{\text{inference}} = \frac{\text{number of correct predictions}}{\text{total number of predictions}} \quad (20)$$

In addition, the mean bias is employed to measure the data utility loss after privacy processing:

$$L_{\text{utility}} = \left| \frac{1}{n} \sum_{i=1}^n x_i - \frac{1}{n} \sum_{i=1}^n \hat{x}_i \right| \quad (21)$$

All experiments are repeated fifty times with different random seeds, and the average results are reported.

5.2 Experimental Results

5.2.1 Inference Risk. Based on the inference accuracies presented in Figure 2, it is evident that different attribute inference models exhibit distinct performance trajectories on the same dataset, suggesting that feature importance varies significantly across inference approaches.

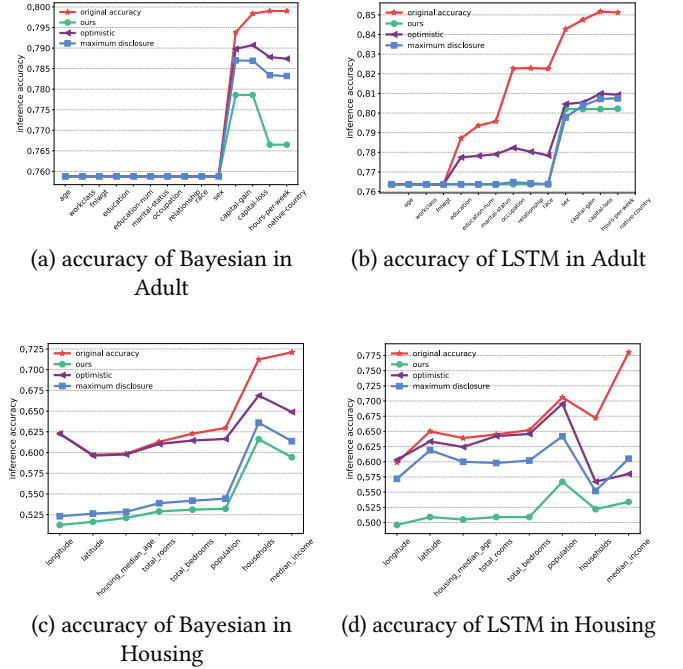


Figure 2: Inference Accuracy with Continual Attribute Sharing

Specifically, the optimistic strategy assumes that each subsequent state transition will yield the highest utility, leading to insufficient privacy protection during the early stages of information sharing and thus higher risks of privacy inference. As information accumulates, the cumulative risk grows, and the privacy protection effectiveness of this strategy deteriorates more rapidly than that of other methods. In contrast, the maximum disclosure algorithm generally outperforms the optimistic one by maximizing instantaneous utility, but it is short-sighted. When cumulative privacy risks emerge later (e.g., Figure 2(b)), its protection converges to the optimistic strategy, losing its early advantage.

¹<https://archive.ics.uci.edu/dataset/2/adult>

²<https://github.com/EngMoElhaggar/California-Housing-Dataset>

By contrast, our proposed approach jointly considers both short-term and long-term utilities, leading to more conservative privacy strategies from the outset to prevent the uncontrolled buildup of privacy risks in later stages. Consequently, it achieves consistently superior privacy protection across early, middle, and late phases of continual information sharing. The effect is particularly evident in the California Housing dataset, where feature–target correlations are stronger, resulting in more substantial suppression of inference risks.

5.2.2 Privacy Strength vs Utility. Privacy inference risk alone cannot fully reflect performance, as privacy protection conflicts with information utility. We thus examine the privacy strength and utility over time, shown in Figure 3.

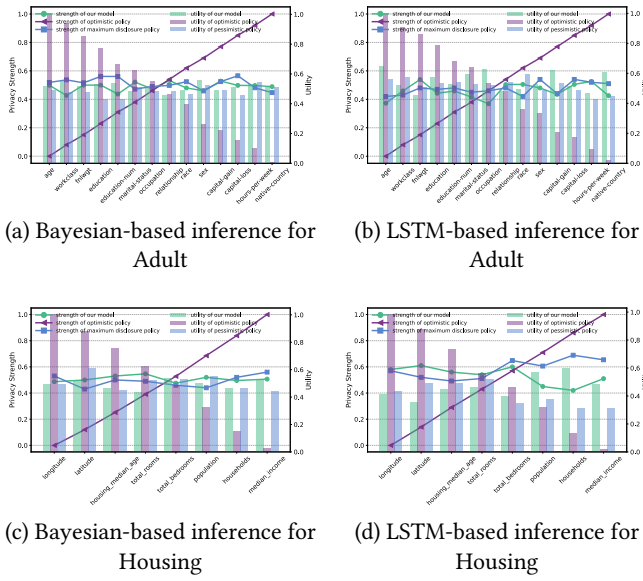


Figure 3: Privacy Strength and Utility for Each Attribute Sharing

From Figure 3, it is clear that the privacy strengths and utilities of both the maximum disclosure algorithm and our proposed method remain relatively stable over time, whereas those of the optimistic algorithm increase approximately linearly with the accumulation of shared information. Although the privacy strengths of our method and the maximum disclosure algorithm are numerically close, our approach generally adopts slightly weaker privacy strengths, particularly in the later stages of sharing, while still achieving lower inference risks.

This observation, in conjunction with the inference risk results in Figure 2, indicates that our method derives more effective privacy strategies by adaptively allocating protection strength. Specifically, it applies stronger protection to information that is more critical for attribute inference, thereby curbing the cumulative growth of privacy risks while preserving greater data utility. Consequently, our approach achieves a more favorable privacy-utility balance, maintaining high-level privacy protection with less degradation in information utility per sharing instance.

5.2.3 Analysis of Utility for User. The effects of privacy protection strength and inference strength are further analyzed and shown in Figure 4.

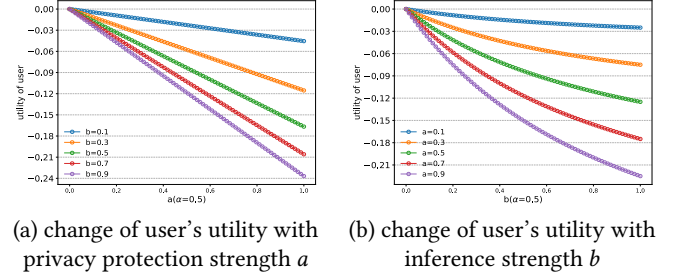


Figure 4: Utility of User with Strength of Privacy Protection and Inference

It is important to note that an increase in inference strength leads to a nonlinear decrease in user utility, whereas an increase in privacy protection strength results in a linear decrease. While stronger privacy protection reduces privacy loss, it simultaneously diminishes the utility of shared information due to the inherent conflict between privacy and utility. In contrast, the rise in inference strength solely affects the privacy leakage experienced by the user, directly exacerbating the decline in utility without altering the information gain.

In summary, the curves shown in Figure 4 demonstrate that our model accurately captures the dynamics of privacy leakage, while also emphasizing that users should consider both the adversary's behavior and the potential loss in information utility when making privacy protection decisions.

6 Conclusion

In this study, we focus on the accumulated privacy risk arising from continual information sharing and propose a Stackelberg game-based model to address this challenge. By modeling the game-theoretic interaction between users and adversaries, we establish the existence of a Stackelberg equilibrium and derive the corresponding equilibrium strategies for both users and adversaries. Furthermore, we develop an approximately optimal method based on Proximal Policy Optimization (PPO) to handle the computational complexity of multi-stage interactions. Experimental results on real-world datasets demonstrate that our model can effectively mitigate accumulated privacy risk by applying higher privacy strength to critical information while preserving more utility for less sensitive data. In future work, we plan to adapt our model to specific real-world scenarios, such as online social networks, to further address accumulated privacy risks in practical applications.

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Appendix

A Proof of Lemma 1

PROOF. Let $\mathbb{B}$ be a compact convex set and $U_{adv}(b; a)$ be continuous with respect to b . According to the Berge Maximum Theorem [3], if $U_{adv}(b; a)$ is compactly continuous with respect to a and continuous with respect to b , then $\arg \max_{b \in \mathbb{B}} U_{adv}(b; a)$ is upper continuous and non-empty in a . Therefore, for any fixed a , the optimal response set $b^*(a)$ exists and is upper-semi-continuous. $\square$

B Proof of Theorem 1

PROOF. According to Lemma 1, the adversary's optimal response $b^*(a)$ to the user's privacy strategy a is non-empty and upper semi-continuous.

We define the Bellman operator $\mathcal{T} : C(\mathbb{S}) \rightarrow C(\mathbb{S})$ as follows:

$$\mathcal{T}V(s) := \max_{a \in \mathbb{A}} \{U_u(s, a, b(s, a)) + \gamma \int V_u(s') P(ds' | s, a, b(s, a))\}$$

Under conditions that utility is continuous and the action space is compact, It follows that the maximum in the Bellman operator is continuous in $s \in \mathbb{S}$, and thus $\mathcal{T}$ maps $C(\mathbb{S})$ to itself.

Further, for any $V_1, V_2 \in C(\mathbb{S})$, according to the Bellman operator, we have

Further, for any $V_1, V_2 \in C(S)$, according to the Bellman operator, we have

$$\begin{aligned} \|\mathcal{T}V_1 - \mathcal{T}V_2\|_\infty &= \sup_{s \in \mathbb{S}} \max_a \{U_u(s, a, b^*(s, a)) \\ &\quad + \gamma E_{s'}[V_1(s')] - \max_a \{U_u(s, a, b^*(s, a)) \\ &\quad + \gamma E_{s'}[V_2(s')]\} \\ &\leq \gamma \sup_{s \in \mathbb{S}} \sup_{a \in \mathbb{A}} |E_{s'}[V_1(s') - V_2(s')]| \\ &\leq \gamma \|V_1 - V_2\|_\infty \end{aligned}$$

There is one and only one fixed-point $V_u^* \in C(\mathbb{S})$ satisfies the Bellman equation based on the Banach fixed point theorem [19]. Therefore, the optimal policy π_u^* can be derived.

Since $b^*(s, a)$ is always the optimal response of adversary and π_u under each state s make the Bellman equation maximum, $\langle \pi_u, \pi_{adv} \rangle$

is the optimal policy combination at each time period, which satisfy the definition of dynamic Stackelberg equilibrium. Meanwhile, the discount is consistent with Bellman operator, i.e., the policies remain optimal in any subsequent subgame. $\square$

C Proof of Theorem 2

PROOF. The utility of the adversary is given by:

$$U_{adv} = -\log \max_{x \in \mathcal{X}} P(x^s) + \log \max_{x \in \mathcal{X}} Q(x^s | \hat{x}_t) - \sum_x p(x) \cdot \log \frac{P(x^s)}{Q(x^s | Z)} - \frac{b}{a_s} F$$

To find the optimal strategy b^* , we compute the first derivative with respect to b :

$$\frac{\partial U_{adv}}{\partial b} = \frac{\lambda e^{-\lambda b/(a+\epsilon)}}{(a+\epsilon)(1-e^{-\lambda b/(a+\epsilon)})} + \frac{\lambda e^{-\lambda b/(a+\epsilon)}}{(a+\epsilon)(1-e^{-\lambda b/(a+\epsilon)})} - \frac{F}{a} = \frac{2\lambda e^{-\lambda b/(a+\epsilon)}}{(a+\epsilon)(1-e^{-\lambda b/(a+\epsilon)})} - \frac{F}{a}$$

The second derivative is:

$$\frac{\partial^2 U_{adv}}{\partial b^2} = -\frac{2\lambda^2 e^{-\frac{\lambda b}{a+\epsilon}}}{(a+\epsilon)^2 \left(1 - e^{-\frac{\lambda b}{a+\epsilon}}\right)^2} < 0$$

indicating that U_{adv} is concave with respect to b , so a unique maximum exists.

Setting $\frac{\partial U_{adv}}{\partial b} = 0$ gives:

$$\frac{2\lambda e^{-\lambda b/(a+\epsilon)}}{(a+\epsilon)(1-e^{-\lambda b/(a+\epsilon)})} = \frac{F}{a}$$

Let $z = e^{-\lambda b/(a+\epsilon)}$, then

$$\frac{2\lambda z}{(a+\epsilon)(1-z)} = \frac{F}{a}$$

and

$$2\lambda a z = F \cdot (a+\epsilon) \cdot (1-z)$$

Solving for z yields:

$$z(a) = \frac{F \cdot (a+\epsilon)}{2\lambda a + F \cdot (a+\epsilon)}$$

That is:

$$e^{-\lambda b/(a+\epsilon)} = \frac{F \cdot (a+\epsilon)}{2\lambda a + F \cdot (a+\epsilon)}$$

Taking the logarithm and solving for b^* gives the optimal inference strategy of the adversary:

$$\begin{aligned} b^* &= -\frac{(a+\epsilon)}{\lambda} \ln \frac{F \cdot (a+\epsilon)}{2\lambda a + F \cdot (a+\epsilon)} \\ &= \frac{(a+\epsilon)}{\lambda} \ln \frac{2\lambda a + F \cdot (a+\epsilon)}{F \cdot (a+\epsilon)} \end{aligned}$$

$\square$

D Proof of Theorem 3

PROOF. The utility of the user is given by:

$$U_u = -\alpha a - \log(1 + e^{-\frac{a}{b+1}})$$

Let $k = \frac{1}{b+1}$, then:

$$U_u = -\alpha a - \log(1 + e^{-ka})$$

The first derivative with respect to a is:

$$\frac{\partial U_u}{\partial a} = -\alpha + \frac{ke^{-ka}}{1 + e^{-ka}}$$

and the second derivative is:

$$\frac{\partial^2 U_u}{\partial a^2} = -k^2 \frac{e^{-ka}}{(1 + e^{-ka})^2} < 0$$

indicating that U_u is strictly concave with respect to a . Therefore, if an interior solution a^* exists, it is the unique global maximum.

Setting $\frac{\partial U_u}{\partial a} = 0$ gives:

$$\alpha = \frac{ke^{-ka}}{1 + e^{-ka}}$$

Let $u = e^{-ka}$, then:

$$\alpha = \frac{ku}{1 + u}$$

and

$$u = \frac{\alpha}{k - \alpha}$$

Substituting back $u = e^{-ka}$, we have:

$$e^{-ka} = \frac{\alpha}{k - \alpha}$$

and

$$a = \frac{1}{k} (\ln(k - \alpha) - \ln \alpha) = (b+1) \left(\ln \left(\frac{1}{b+1} - \alpha \right) - \ln \alpha \right)$$

Therefore, when $\frac{1}{b+1} - \alpha > 0$, a has positive real solutions, i.e., $a^* = a$. Otherwise, it indicates that U_u is non-monotonically increasing with respect to a . Thus, the optimal a lies on the boundary of the feasible region. Typically, if $a \geq 0$, the optimal choice is $a = 0$. In other words, when $\frac{1}{b+1} \leq \alpha$, no internal fixed points should be expected, and the user will select the minimum protection strength to maximize U_u . $\square$